Initial Project Planning Template

Date	25 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Problem Definition	USN-1	As a team, we identify and define the agricultural problem that OptiCrop aims to solve	2	High	Team	26-06-2026	02-07-2026
Sprint-1	Problem Definition	USN-2	As a team, we gather and analyze business requirements for the crop recommendation system	2	High	Team	26-06-2026	02-07-2026
Sprint-1	Problem Definition	USN-3	As a researcher, I conduct a literature survey on existing crop recommendation ML techniques	3	High	Team	27-06-2026	02-07-2026

Sprint-1	Problem Definition	USN-4	As a team, we analyze the social and business impact of OptiCrop	1	Medium	Team	27-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-5	As a data analyst, I download the Crop_recommendation.csv dataset from Kaggle	1	High	Team	27-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-6	As a developer, I import and set up required Python libraries (NumPy, Pandas, Scikit-learn)	1	High	Team	27-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-7	As a data analyst, I read and explore the dataset structure and features	2	High	Team	28-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-8	As a data analyst, I perform univariate analysis on soil and environmental features	2	Medium	Team	28-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-9	As a data analyst, I conduct bivariate analysis (humidity vs crop labels)	2	Medium	Team	28-06-2026	02-07-2026
Sprint-2	Epic 2: Data Collection & Analysis	USN-10	As a data analyst, I perform multivariate analysis across all agricultural parameters	2	Medium	Team	28-06-2026	02-07-2026
Sprint-3	Epic 3: Data Preprocessin g	USN-11	As a data engineer, I check and handle null/missing values in the dataset	1	High	Team	28-06-2026	02-07-2026

Sprint-3	Epic 3: Data Preprocessing	USN-12	As a data engineer, I detect and treat outliers using IQR and log transformation	2	High	Team	28-06-2026	02-07-2026
Sprint-3	Epic 3: Data Preprocessing	USN-13	As a data engineer, I extract seasonal crop information for feature engineering	2	Medium	Team	28-06-2026	02-07-2026
Sprint-3	Epic 3: Data Preprocessing	USN-14	As a data engineer, I split the dataset into training and testing sets using train_test_split	1	High	Team	29-06-2026	02-07-2026
Sprint-4	Epic 4: Model Building	USN-15	As an ML engineer, I implement K-Means Clustering with Elbow Method	3	High	Team	30-06-2026	02-07-2026
Sprint-4	Epic 4: Model Building	USN-16	As an ML engineer, I train a Logistic Regression model for crop prediction	2	High	Team	30-06-2026	02-07-2026
Sprint-4	Epic 4: Model Building	USN-17	As an ML engineer, I evaluate model performance (accuracy, precision, recall, F1, confusion)	2	High	Team	30-06-2026	02-07-2026
Sprint-4	Epic 4: Model Building	USN-18	As an ML engineer, I save the best model using Pickle and test crop predictions	2	High	Team	30-06-2026	02-07-2026
Sprint-5	Epic 5: Application Building	USN-19	As a frontend developer, I design HTML pages (Home, About, FindYourCrop) with Bootstrap	3	High	Team	30-06-2026	02-07-2026
Sprint-5	Epic 5: Application Building	USN-20	As a backend developer, I build Flask routes and integrate the ML model for predictions	3	High	Team	30-06-2026	02-07-2026

Sprint-5	Epic 5: Application Building	USN-21	As a full-stack developer, I implement the "Previous Crops" feature to save and delete history	2	Medium	Team	01-07-2026	02-07-2026
Sprint-5	Epic 5: Application Building	USN-22	As a tester, I run and validate the application for accurate crop recommendations and history tracking	2	High	Team	02-07-2026	02-07-2026
Sprint-5	Epic 5: Application Building	USN-23	As a team, we deploy the application locally and prepare for final demonstration	2	High	Team	02-07-2026	02-07-2026